ASHLEY BELLAS-MANLEY

ADDRESS	Department of Aerospace Engineering Sciences University of Colorado Boulder 3775 Discovery Drive, Boulder, CO, 80309
EMAIL & TELEPHONE	ashley.bellas@colorado.edu; 720-900-9363

EDUCATION

08/2014-05/2021	University of Colorado Boulder, Department of Physics Ph.D. in Geophysics: *Reconciling the Rheology of Earth's Lithosphere Across Vastly Different Length- and Time-Scales* Thesis advisor: Prof. Shijie Zhong
09/2009-05/2014	University of British Columbia B.Sc. in Geophysics

ACADEMIC APPOINTMENTS

08/2024-present	Professional Research Associate Department of Aerospace Engineering Sciences, University of Colorado Boulder
08/2022-08/2024	Postdoctoral Associate Department of Aerospace Engineering Sciences, University of Colorado Boulder
08/2021-08/2022	Postdoctoral Fellow Department of Earth, Atmospheric and Planetary Sciences, Massachusetts Institute of Technology
05/2021-08/2021	Postdoctoral Associate Department of Physics, University of Colorado Boulder

PEER-REVIEWED PUBLICATIONS

1. **Bellas-Manley, A.,** Kang, K., Becker, T. W., & Nerem, R. S. (2026). 3D variations in viscosity reconcile the strength of the lower mantle inferred from glacial isostatic adjustment and mantle convection. *Geophysical Research Letters*, 53, e2025GL120160. https://doi.org/10.1029/2025GL120160
2. Thompson PR, M.J. Widlansky, B. Beckley, **A. Bellas-Manley**, D.P. Chambers, B.D. Hamlington, S. Jevrejeva, F.W. Landerer, E. Leuliette, M.A. Merrifield, G.T. Mitchum, R.S. Nerem, and W. Sweet, "Sea Level Variability and Change [in State of the Climate 2024]." Bulletin of the American Meteorological Society. 106 (8) (August, 2025): S200, https://doi.org/10.1175/2025BAMSStateoftheClimate.1
3. **Bellas-Manley, A.,** Nerem, R. S., & Hamlington, B. D. (2025). Extrapolation of the satellite altimeter record to understand regional variations in future sea level change. Journal of Geophysical Research: Oceans, 130, e2024JC022094. https://doi.org/10.1029/2024JC022094
4. Karnauskas, K. B., Nerem, R. S., Fasullo, J. T., **Bellas-Manley, A.,** Thompson, P. R., Coats, S., et al. (2025). Diagnosing regional sea level change over the altimeter era. Journal of Geophysical Research: Oceans, 130, e2024JC022100. https://doi.org/10.1029/2024JC022100
5. Loomis, B. D., Sabaka, T. J., Rachlin, K.E., Croteau, M. J., Lemoine, F. G., Nerem, R. S., & **Bellas-Manley, A.** (2025). Optimized J2 recovery for multi-decadal geophysical studies. Geophysical Research Letters, 52, e2024GL114472. https://doi.org/10.1029/2024GL114472

6. Rodell, M., Barnoud, A., Robertson, F.R., Richard P. Allan, **A. Bellas-Manley**, M. G. Bosilovich, D. Chambers, F. Landerer, B. Loomis, R. S. Nerem, M. M. O'Neill, D. Wiese & S. I. Seneviratne (2024), An Abrupt Decline in Global Terrestrial Water Storage and Its Relationship with Sea Level Change. *Surv Geophys*. https://doi.org/10.1007/s10712-024-09860-w
7. Hamlington, B.D., **Bellas-Manley, A.,** Willis, J.K. *et al*. (2024), The rate of global sea level rise doubled during the past three decades. *Nature Commun Earth Environ* 5, 601. https://doi.org/10.1038/s43247-024-01761-5
8. **Bellas-Manley, A.** & L. Royden (2024), Basal Mantle Flow Over LLSVPs Explains Differences in Pacific and Indo-Atlantic Hotspot Motions, *J. Geophys. Res.: Solid Earth,* 129, e2023JB027636. https://doi.org/10.1029/2023JB027636
9. **Bellas, A.,** S.J. Zhong, A.B. Watts (2022), Reconciling lithospheric rheology between laboratory experiments, field observations and different tectonic settings, *Geophysical Journal International*, Volume 228, Issue 2, Pages 857–875, https://doi.org/10.1093/gji/ggab382
10. **Bellas, A.,** & Zhong, S. (2021). Effects of a weak lower crust on the flexure of continental lithosphere. *Journal of Geophysical Research: Solid Earth*, 126, e2021JB022678. https://doi.org/10.1029/2021JB022678
11. **Bellas, A.,** & Zhong, S. J. (2021). Seismic strain rate and flexure at the Hawaiian Islands constrain the frictional coefficient. *Geochemistry, Geophysics, Geosystems*, 22, e2020GC009547. https://doi.org/10.1029/2020GC009547
12. **Bellas, A.,** S.J. Zhong, & A.B. Watts (2020). Constraints on the rheology of the lithosphere from flexure of the Pacific Plate at the Hawaiian Islands. *Geochemistry, Geophysics, Geosystems*, *21*, e2019GC008819. https://doi.org/10.1029/2019GC008819.
13. **Bellas, A.,** S.J. Zhong, D. Bercovici, E. Mulyukova (2018), Dynamic weakening with grain-damage and implications for slab detachment, *Physics of the Earth and Planetary Interiors*, Volume 285, Pages 76-90, ISSN 0031-9201, https://doi.org/10.1016/j.pepi.2018.09.001.

INVITED TALKS

Harvard Biweekly Seminar in Planetary Physics, Cambridge, MA, April 1st 2026. A. Bellas Manley, *Can 3D variations in viscosity reconcile the strength of the lower mantle inferred from GIA and mantle convection?*

RESEARCH FUNDING & PROPOSALS

2024–2028 — *Understanding Future Sea Level Change by Combining NASA Satellite Observations and Climate Models Using Machine Learning.* NASA Sea Level Change Science Team. **Co-Investigator; funded.** Total award: $984,000.

2026–2029 — *3D Earth Structure in GIA Models and Impacts on Satellite Observations of a Dynamic Planet.* NASA Early Career Investigator Program. **Principal Investigator; pending.** Amount requested: $293,453.

2027–2028 — *From GRACE to Water Decisions: Co-designing Multi-Sensor, AI-Enabled Water Storage Monitoring for the Western United States.* NASA Accelerating Earth Solutions. **Principal Investigator; pending.** Amount requested: $149,621.

CONFERENCE PRESENTATIONS

Chapman Conference on Usable Sea Level Projections, Montreal, QC, June 7-11 2026. *Putting FACTS in the altimeter reference frame: probabilistic sea level projections in terms of geocentric sea level.*

International Association of Geodesy Scientific Assembly, Rimini, Italy, Sept. 1-5, 2025. *Satellite Geodesy for Assessing Future Sea Level Change Impacts.*

NASA Solid Earth Team 2.0 Meeting (2024), Washington, D.C., *The Half-Century Record of Changes in the Earth's Oblateness from Satellite Laser Ranging: What Is It Telling Us?*

CESM2 Workshop (2024), Boulder, CO, *Using the CESM2 Large Ensemble to Evaluate CSEOF Separation of Internal and Forced Components of Sea Level Change*

Colorado Glaciology Conference (2024), Boulder, CO, *Earth's Oblateness as a Long Term Record of the Cryosphere*

AGU Fall Meeting (2023), San Francisco, CA. *Data-Driven Approaches to Understanding Future Regional Sea Level Change*, G53B-03.

NASA Ocean Surface Topography Science Team Meeting (2023), San Juan, Puerto Rico. *Impacts of GIA Modeling Uncertainties on the Closure of the GMSL Budget, 10.24400/527896/a03-2023.3824*

NASA GRACE-FO Science Team Meeting (2023), Boulder, CO. *Impacts of GIA Modeling Uncertainties on the Closure of the Global Mean Ocean Mass Budget*

NASA Sea Level Change Science Team Meeting (2023), Pasadena, CA. *Data-Driven Approaches to Understanding Regional Variations in Future Sea Level Change*

Study of the Earth's Deep Interior Conference (2022), Zurich, Switzerland. *Basal Mantle Flow Over LLSVPs Explains Differences in Pacific and Indo-Atlantic Hotspot Motions*

AGU Fall Meeting (2021), New Orleans, LA. *Effects of a Weak Lower Crust on the Flexure of Continental Lithosphere*, T11D-05.

AGU Fall Meeting (2021), New Orleans, LA. *Reconciling Lithospheric Rheology Between Laboratory Experiments, Field Observations, and Different Tectonic Settings*, MR43A-06.

AGU Fall Meeting (2020). *Testing the Yield-Stress Envelope Method Against Finite Element Models of Flexure*, T011-0008.

AGU Fall Meeting (2019) San Francisco, CA. *Constraining the Frictional Coefficient: a Comparison of Strain Rate Inferred from Seismicity and 3D Viscoelastic Loading Models at Hawaii*, MR44A-03.

AGU Fall Meeting (2019) San Francisco, CA. *Elastic Thickness: A Comparison of Estimates from Fully Dynamic Viscoelastic Models and the Yield-Strength Envelope Method*, MR51B-0040.

Gordon Research Conference (2019) Holyoke, MA. *Constraining the rheology of the lithosphere using flexure at the Hawaiian Islands.*

AGU Fall Meeting (2018) Washington, D.C. *Constraining mantle rheology at lithospheric conditions using observations of flexure at the Hawaiian Islands*, MR24A-01.

Study of the Earth's Deep Interior Conference (2018), Edmonton, AB, Canada. *Dynamics of a Subducted Slab with Grain-Damage*

TEACHING

Guest Lecturer, Satellite Geodesy (ASEN 6070), CU Boulder — 2024
Guest Lecturer, Remote Sensing Data Analysis (ASEN 6337), CU Boulder — 2024
Course Facilitator, Remote Sensing Seminar (ASEN 5210), CU Boulder — 2022
Teaching Assistant, Introduction to Earth and Planetary Geophysics (12.001), MIT — 2022
Teaching Assistant, General Physics I & II (PHYS 1110/1120), CU Boulder — 2014–2018
Led undergraduate recitation sections and assisted with course assessment.

SERVICE

Peer Reviewer: *Science* (2022); *Nature Sustainability* (2026); *Communications Earth & Environment* (2024); *Geophysical Journal International* (2026); *Geophysical Research Letters* (2023); *Journal of Geophysical Research: Solid Earth* (2020, 2021, 2024); *Geochemistry, Geophysics, Geosystems* (2020, 2022).